



PRODUCT NAME

Industrial-strength Powder Detergent Formula

Description

laundry powder detergent

Technical Specifications

pH Level:	10
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Sodium Carbonate	Sodium Carbonate	35%	Water softener and pH buffer
2	Sodium Sulfate	Sodium Sulfate	20%	Filler and flow aid
3	Sodium Tripolyphosphate	Sodium Tripolyphosphate	15%	Builder and sequestering agent
4	Sodium Dodecylbenzenesulfonate	Sodium Dodecylbenzenesulfonate	12%	Anionic surfactant
5	Sodium Percarbonate	Sodium Percarbonate	10%	Oxidizing agent and stain remover
6	Zeolite A	Zeolite	5%	Detergency booster and water softener
7	Enzymatic Complex	Protease, Amylase	2%	Stain removal and fabric care
8	Optical Brightener	Disodium Distyrylbiphenyl Disulfonate	0.5%	Whitening agent
9	Fragrance Powder	Parfum	0.5%	Fragrance
Total			100%	

Manufacturing Process

Phase 1: Manufacturing Phase

Phase 2: Manufacturing Phase

Phase 3: Manufacturing Phase

Phase 4: Manufacturing Phase

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions